

Trans-impedance Amplifier (TIA) Design for Visible Light Communication (VLC) using Commercially Available OP-AMP

Syifaul Fuada¹, Angga Pratama Putra², Yulian Aska³, Trio Adiono⁴

Microelectronic Center - Pusat Penelitian Mikroelektronika, VLC Research Group ITB

PAU Building 4th Floor, ITB campus, Tamansari street No. 126, (ZIP: 40132)

Telp: +62-22-2506280/Fax: +62-22-2508763, Bandung city, West Java, INDONESIA

{syifaulfuada¹, anggapatap², yulian.aska³}@pme.itb.ac.id, Tadio@stei.itb.ac.id⁴

Abstract—Trans-impedance amplifier (TIA) are widely used for optical sensing applications (precision instrumentation) and optical communication, such as: fiber optic, IR communication and also Visible Light Communication (VLC). TIA convert the current (I) into voltage (V). Application TIA design depend of objectives requirement system, for high speed as well as high-band, low noise, low power and others. This paper explains the empirical experiment of **TIA for high-band VLC with over 1-MHz Gain Bandwidth Product (GPWP)** estimated. We implement general topology mode: as photovoltaic. The selection of the best resistor feedback (Rf) and compensator (Cf) are presented, we compare simulation approach against real implementation. The OP-AMP chosen from Texas Instruments® product, OPA 656. The TINA™ SPICE® simulator used as simulation's tool. This research contribution lies in applied of engineering design methodology with detailed step to develop major VLC system, we focus on photodiode amplifier only. By depth analysis, can be known next research addresses and obtain the best results without trial and error methods.

Keywords— OPA656; Trans-impedance Amplifier (TIA); Visible Light Communications (VLC)

I. INTRODUCTION

Visible Light Communication (VLC), also known as **Li-Fi**, is new technology and an alternative method for wireless communication besides RF and IR. Research topic about VLC became an interesting studies for several decades forward because there are any open problem need to solve. VLC studies not only be done by companies but also academicians at the higher educations of the world.

Based on literature studies, exploration about development VLC can be divided into at least ten research focuses, i.e. **1)** communication system (e.g. bidirectional, multiuser, MIMO, bidirectional-multiuser, etc.); **2)** channel modelling (e.g. optimal placement, propagation model, etc.) and its improvement, **3)** proposed an algorithm and its compared, **4)** modulation model, **5)** Analog Front-end model, **6)** Digital Signal Processing (DSP) optimization which cover System on Chip (SOC) design, **7)** development of material device for LED and photo-detector future; **8)** customizable VLC block into digital-analog ICs, **9)** integration VLC with

other communication such as Power Line Carrier (PLC), Radio Frequency (RF), **10)** applied in education as project based learning materials and etc.

According to the VLC researches which even recorded till now, an Analog Front-End (AFE) is one of VLC part platform which could be explored more, this part has an influence in order to produce high Signal-Noise-Ratio (SNR or S/N). Generally, an AFE circuit of VLC system divided into two parts, i.e. transmitter and receiver. The transmitter consist of analog/digital driver circuit, and **the receiver consist of photo-detector (photodiode, phototransistor, solar cell, Light Dependent Resistor), TIA and specific analog circuit (amplifiers, filters and other circuit configurations)**. However, several researchers replace digital ADC/DAC with analog, then implemented and tested into VLC system with aim to compare speed, computation requirement and implementation cost between digital-domain against analog- domain. Hence, ADC and DAC be included into AFE board.

Trans-impedance Amplifier (TIA) or another name called as photodiode amplifier, has a function for converting the photodiode photocurrent output (I_{sc}) to a voltage signal (V_o). TIA has been implemented in various optic application, not only for optical sensing (measurement purpose) but also for optical communication (i.e. IR, VLC, fiber-optic). They are can be built by discrete circuit mechanism, such as: [1], [2], or by MOS transistor circuitry. Its circuit design divided into several aspects, such as: precision instrumentation TIA, high speed TIA, low noise TIA [3], gain stability TIA [4], manual programmable TIA [21] and other related objectives.

Applying TIA based MOS transistor at VLC system tend to be more complicated than discrete component, but we can create a design accordance with the research goals, not only for practically implemented in specific purpose but also it is more advanced performance than specification of available OP-AMP by now, such as [5], [6], [7], [8]. In other hand, TIA based discrete component are major in use because light weight implementation, lower effort, more inexpensive and presumably been capable enough to show perform well.

Most of researches about TIA are describes about detailed in design and implementation for precision measurement, there are still rarely research which cover the

TIA design for VLP applications. Thus, in this paper, a case study is presented based experiment specifically on implementing OP-AMP circuit. We implement general topology of TIA, then calculate all of usage component with three approaches: mathematically calculation, simulation, and real experiment. Hope this research report became as our reference to build complete system further, AFE in particular.

We obtain higher noise at TIA if targeting low power and high speed, therefore need special requirement. It can take by select specific OP-AMP and advanced electronic configuration circuitry. In this paper, we focus on analyze high-band TIA only. A Low noise design would be explained more in detail in another occasions [13].

This paper is divided into five sections. First is discusses about background and research target. Second is discusses about our VLC platform as main reference, it is followed by our TIA operation mode and component specification (OP-AMP). Fourth is discusses thoroughly about the evaluation results by simulation using and real experiment, this parts is design functional test. Last parts are conclusions and references, respectively.

II. SYSTEMS DESCRIPTION

A. VLC Systems

Fig. 1 depicts a block diagram of VLC environment. We use an analog modulation, DC-OFDM and ACO-OFDM. A Detailed explanation of those block was presented at [9] and [10]. VLC designed by point to point with one link (download) with >1 MHz bandwidth target estimated. Since OFDM as multicarrier signal, we assume the system can transmit over 10Mb/s. In order to achieve this target and received signal in receiver (Rx) must be processed by low error (correctly), an available commercially analog component is chosen providently. We use high brightness white LED which has high FOV for wide light distribution. Photodiode selected as photo-detector, because has fall-rise time better than photovoltaic, photo-transistor and Light Dependent Resistor (LDR).

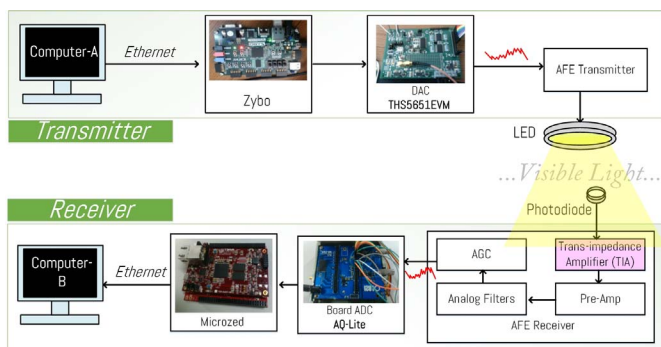


Fig. 1. VLC environment which used in this research

Basic principle described below: initially, an FPGA at transmitter (T_x) sent data from Computer A and emitted by white color LED. Transmitted data will be received by a

photodiode and its output current converted into voltage. Photodiode generates small current that is proportional with light illumination level. A high gain across the TIA is required in order to overcome weak photocurrent signals when the receiver (R_x) be kept out from light source.

However, increasingly large gains, the bandwidth will be narrower. In this case, adding adjustable gain by manually (use variable resistor) as well as an automatic gain adjuster at TIA is not main option. Select a photodiode with high current output typical not an easy matter, because susceptible with high-noise. Possible approaches is adding an amplifier circuit. Of course it will be paid by more complexity in packaging circuitry. Amplifier circuit connected to analog filters, a circuit for eliminating noise signal which comes from other lights interferences. Then, data from AFE processed by FPGA at R_x and displayed in computer-B.

Discussions in this paper are limited focus about Trans-impedance Amplifier only. They are indicated as pink-boxes one in AFE section.

B. Photodiode Mode Operations and Equivalent

Photodiode has two operation modes, i.e. photoconductive and photovoltaic, each mode has its advantages and disadvantages. Photovoltaic-mode has several characteristics such as: a) without bias, b) non-generate dark current, c) linear, d) low noise, and e) precision instrumentation compatible. Whereas photoconductive-mode are: a) reverse bias, b) generate dark current, c) nonlinear characteristic, d) has large-noise, and e) high speed compatible. These configuration mode are illustrated in Fig 2.

We use the photovoltaic mode in this experiment, because it is low-noise and precision characteristics. Therefore, an information signal emitted from LED can be received with minimum noise level without any adding specific circuitry for minimizing noises. The high-speed, as main target, can be obtained by select an OP-AMP with high-band type.

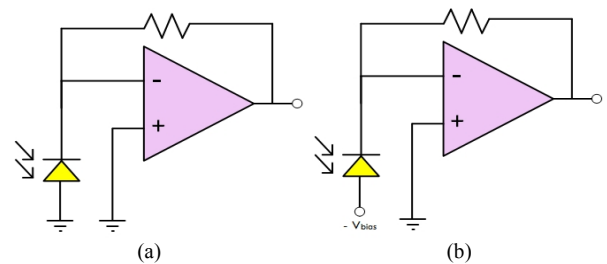


Fig. 2. TIA models for photodiode applications: (a) photovoltaic mode; (b) photoconductive mode

C. OP-AMP Specifications

All of the operational amplifiers used in this research are manufactured from Texas Instrument (TI) Inc. There are three main reasons for using products from TI Inc., the SPICE® circuit files that available for most of their products, the sampling option they offer and the availability of specific simulations tool to prove the design, in this term we use TINA™ software.

TABLE I. SELECTED OP-AMP PARAMETERS

Parameters	Values
OP-AMP type	OPA 656
Input Capacitance (C_{in}) from ($C_{diff} + C_{comm}$)	3.5 pF
Rise and Fall Time	1.5 ns
Gain Bandwidth Product (GBWP)	23 MHz ($G = +10$)
Voltage Noise Density (e_n)	7nV/ $\sqrt{Hz}$ ($>100KHz$)
Current Noise Density (i_n)	1.3fA/ $\sqrt{Hz}$ ($>100KHz$)
Slew Rate	290 V/us
OP-AMP saturation ($R_L = 100\Omega$)	+3.5 V and -3.5V

Table 2 shows key specifications for the OPA656, whose has unity-gain bandwidth of 500 MHz at gain = 1, low input bias current, low offset and drift, low distortion, high output current and low input voltage noise make it a great fit for high speed photodiode amplifier in VLC applications. This OP-AMP type has been widely implemented at various applications, such as for pre-amp [14]. Research [15] investigate based simulation approach, however, we will be performed by experiments.

We must select GBWP at least 50% - 60% of the value specified of the op amp's data sheet [11]. However, we obtain about ~5 MHz only, a 5% of GBWP from 500MHz (theory).

D. Gain and Capacitor Feedback Calculations

Based on the SP8-ML KODENSHI photodiode datasheet, maximum short current (I_{sc}) 330 μ A [12]. However, based real experiment, the I_{sc} max. is about 24.1 μ A. The saturated output voltage value of TIA must low voltage, we have permitted about 1V (not require high gain). Therefore, R_f value are given by eq.1, we obtain resistor 47K Ω as standard available resistor.

$$R_f = \frac{V_{out_{max.}} - V_{out_{min.}}}{i_{photodiode_{(max.)}}} = \frac{1V - 0V}{24\mu A} = 41.7\Omega (\sim 47K) \quad (1)$$

The value of C_f with a 45 degrees phase margin is given by eq.2, the maximum value of C_f can be determined from the desired bandwidth is 10pF, we ensure which our circuit meet the 1 MHz of stated bandwidth.

$$C_f = \sqrt{\frac{C_i + C_{in}}{2\pi * R_f * f_{GBWP}}} = 2.16 \text{ pF} \quad (2)$$

E. Amplifier Gain Bandwidth Calculations

We obtain the value C total by summing $C_i + C_{in} + C_f = (3.4 + 3.5 + 2.16)$ pF, where C_i is terminal capacitance of the photodiode then C_{in} is input capacitance total of the OP-AMP and then C_f is capacitor feedback as calculated as eq.2. Therefore we obtain $C_{tot} = 9.06$ pF. The min. specification of OP-AMP used is following the eq.3 until eq.6.

Based on eq.6 we obtain GBWP which the unity bandwidth of the OP-AMP. For stability design, the rule must by eq.5. Finally we obtain $GBWP > 4.111$ MHz and the OPA656 is suitable for choose with 5 MHz GPWP.

$$f_l = \frac{C_f}{(C_{in} + C_t)} (GBWP) \quad (3)$$

$$f_p = \frac{1}{2\pi C_f R_f} \quad (4)$$

$$f_l > f_p \quad (5)$$

We obtain 1.565 MHz < 1.568 MHz

$$f_{GBWP} > \frac{C_{in} + C_f}{(2\pi * R_f * C_f^2)} \quad (6)$$

$$f_{-3dB} = \sqrt{\frac{f_{GBWP}}{2\pi * R_f * (C_t + C_{in})}} = 1.567 \text{ MHz} \quad (6)$$

III. EXPERIMENT RESULTS

A. Simulation Result Against Real Experiment Data

To ensure the correctness design, we use simulation and real experiment then compare both. Actually, the typical characteristic of OP-AMP, such as: open-loop gain, phase margin, amplifier pulse response, etc. was informed at datasheet, however, self-experiment be more convince in order to get the best systems.

Figure 3 is the TINA-TI™ simulation schematic of the TIA, this software is used to verify circuit performance. Current source i photodiodes and 3.4pF $\rightarrow$ 1.7 capacitor C_t form a simple simulation model for the photodiode.

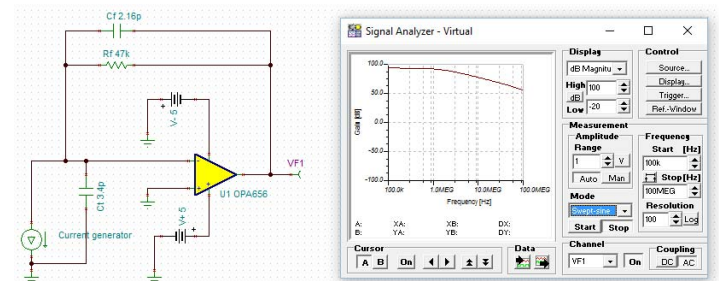


Fig. 3. TIA simulation using TINA™

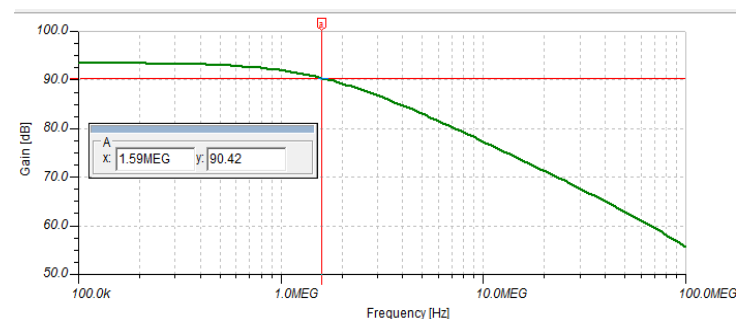


Fig. 4. AC transfer function of TIA simulation, cut-off frequency = 1.59MHz

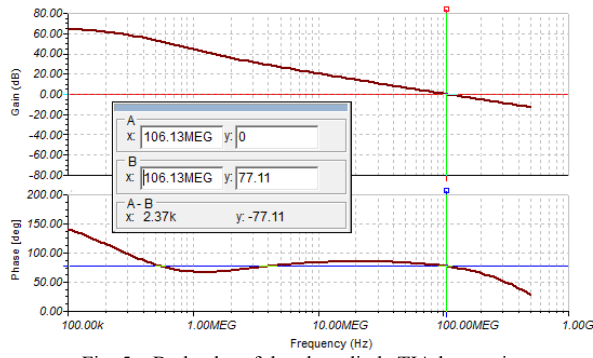


Fig. 5. Bode plot of the photodiode TIA loop gain

The AC transfer function was measured using an AC transfer characteristic analysis in -3dB point was 1.59 MHz (simulated) and 1.50 MHz (measured) the loop gain plot shows the loop closure point, loop gain is 0dB at 106.13 MHz with 77.11 degrees of phase margin. Table 2 is compare results about f-3dB and phase margin, and Table 3 is compare result between design's expectation and reality.

TABLE II. COMPARE PARAMETERS

Parameter	Calculation	Simulation	Real experiment
Frequency cutoff	1.567 MHz	1.59 MHz	1.50 MHz
Phase margin	>45 degrees	77.11 degrees	-

TABLE III. COMPARISON OF DESIGN GOALS WITH SIMULATED AND MEASURED PERFORMANCE (R=47KΩ)

Parameter	Goal	Simulation	Real experiment
$V_{out} (0 \mu A)$	0.2V	0.23245V	0.2008V
$V_{out} (18 \mu A)$	1V	1.08	0.9

B. TIA Characteristic

The experimental scheme to describe the characteristics of a TIA that used in research, involved tools and materials, such as: HAMEG® programmable power supply HMP4030, Coochip® LED module, photodiode SP8-ML and digital multi-meter FLUKE® 15B+. Experimental set-up shown in Fig 6. An OPA656 is SOIC OP-AMP, therefore in order to be placed at breadboard, we use SOIC to DIP adapters which created by self-design. The output from power supply is swept from 8V to 14V (0.25V per step) with current of 1 Ampere, so the LED power can varied. The LED power will affect the illumination level which emitted by LED. A TIA circuit placed confront near LED with constant distance, at 50cm. Then achieved data is plotted by EXCELL™.

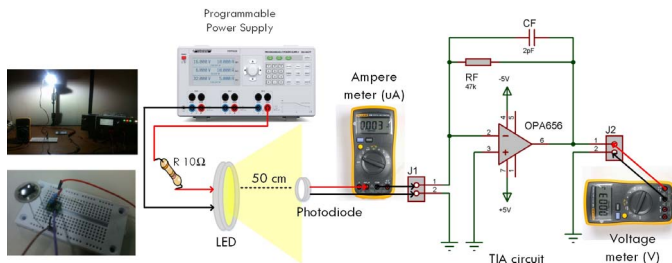


Fig. 6. Experimental set-up

At dark condition, the output voltage measured is 200.8 mV, then simulated output voltage is 233.45 mV. This small current comes from ambient light. In [19] propose an approach to find real output voltage of TIA, by $V_o = I_{inverting} + I_{ambient-light}$.

According to calculation, the ideal Rf value is about 47KΩ. As a comparison, we use different Rf values with three resistors, smaller (15K & 33K) and bigger (56K and 100K) as parameters. The results show that by the greater gain, the output voltage will increases and the current flow also differ. However, ideally, Rf value which near with teoretically calculation is more recommended to obtain an optimal bandwidth. Thus shouldn't choose higher random gain, because greater gain, the output voltage will be saturated and lead clipped signal. Furthermore, information signal couldn't processed correctly.

By knowing the ability of OP-AMP gain, we can design amplifier circuit accurately. These steps is a standard approaches methodology. It was ever undertaken by [18] in making pre-amp for VLC applications.

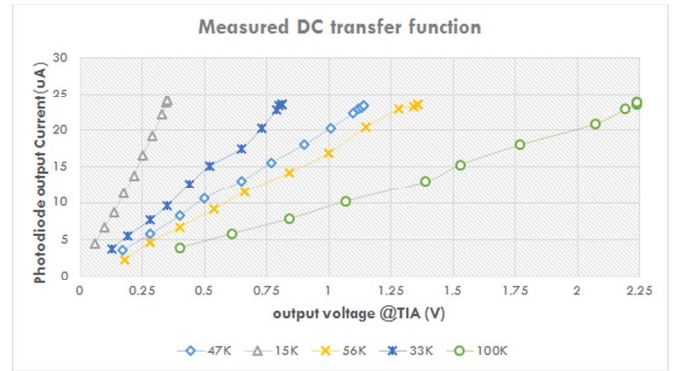


Fig. 7. Measured DC transfer function of the TIA with varied resistor: 15KΩ, 33KΩ, 47KΩ, 56KΩ and 100KΩ

Next, we conduct experiment for knowing correlation between LED power against photocurrent and output voltage. The experiment scheme is same with Fig. 6. The results show that by the greater output power of LED, the lumen (lx) will be higher, then photocurrent and output voltage will be increases. Because photocurrent is proportional to the LED optical power [20]. However, in particular led emission brightness levels, they will be saturated.

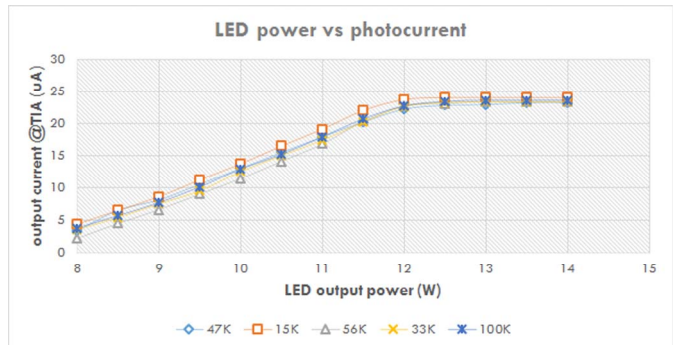


Fig. 8. LED power against photodiode current with varied resistor: 15KΩ, 33KΩ, 47KΩ, 56KΩ and 100KΩ

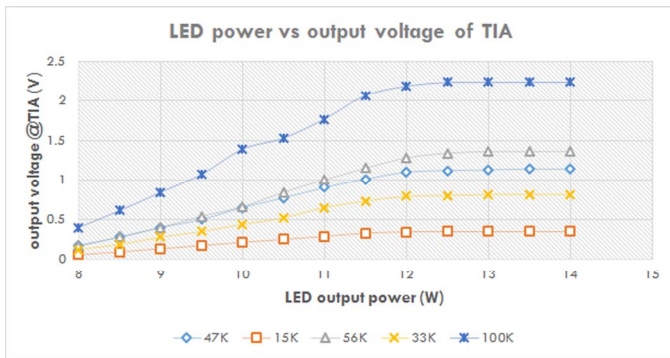


Fig. 9. LED power against output voltage of TIA with varied resistor: 15K Ω , 33K Ω , 47K Ω , 56K Ω and 100K Ω

C. Recommended PCB design

For lay-outing the TIA's PCB must be carefully, because incorrect in designing can make circuits are unstable because parasitic inductance and capacitance generated at high frequency by the PCB traces themselves factor [16]. A detailed tutorial was presented at [17]. TIA board will be fabricated in a double layer PCB with discrete component. To obtain a voltage stability, we use dual power supply +5 and -5V. The fixed layout is shown in Fig.10, a SMA socket is used as connector from TIA to an amplifier circuit stage.

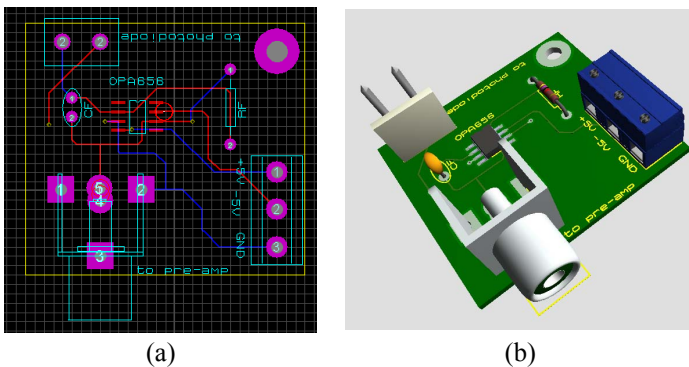


Fig. 10. PCB layout of the TIA which designed by PROTEUS®:

IV. CONCLUSION

TIA has an important and a substantial role in optic communication, the function is converter device, photocurrent to voltage. Emitted data(s) from the LED, as transmitter device, must be read correctly by photodiode. Thus precision design of TIA is needed and also require the high-band width to support high speed communications. In this paper, we calculate a TIA circuit for Visible Light Communications (VLC) with general topology design. The results show that the mathematically calculation, about frequency response is same with simulations results and real experiment. We have been done analyze bode plot and DC transfer response.

For future works, we will be fabricated the TIA module, integrated with other system and demonstrated. Varied distances test between LED to TIA will be involved. We are planned to design multi-TIA with various gain.

- [1] D. Wen, W. Cai, Y. Pan, "Design of Underwater Optical Communication System," OCEANS Shanghai, April 2016.
- [2] P. T. Patil, G.K. Mukherjee, A.K. Sharma, R.R. Mudholkar, "High-Gain Transimpedance Amplifier (TIA) for Night Airglow Photometer," Int. J. of Electronic Engineering Research RIP, Vol. 1(2), pp. 109–116, 2009.
- [3] T-Y. Lin, R.J. Green, P.B. O'connor, "A low noise single-transistor transimpedance preamplifier for Fourier-transform mass spectrometry using a T feedback network," Review of Scientific Instruments, Vol. 83, 2012. DOI: 10.1063/1.4751851.
- [4] S. Deng, E. Cleary, M.AP McAuliffe, L. Lewis, "Design of an auto-gain control transimpedance amplifier for optical sensing applications," Proc. of the 27th Irish Signal and Systems Conf. (ISCC), June 2016. DOI:
- [5] E. Kamrani, *et al.*, "Low-Noise, High-Gain Transimpedance Amplifier Integrated with SiAPD for Low-Intensity Near-Infrared Light Detection," IEEE Sensors and J., Vol. 4(2), pp. 248-269, January 2014.
- [6] R.Y. Chen, Z-Y. Yang, "CMOS Transimpedance Amplifier for Gigabit-per-Second Optical Wireless Communications," IEEE Transaction on Circuits and Systems, Vol. 63(5), pp. 418-422, May 2016.
- [7] B. Moeneclaey, *et al.*, "A 40-Gb/s Transimpedance Amplifier for Optical Links," IEEE Photonics Technology Letters, Vol. 27(13), pp.1375-1378, July 2015.
- [8] X. Kuang, F. Fan, T. Wang, "A Transimpedance Amplifier with DC Photocurrent Rejection for Infrared Optical Receiver," Proc. of the Int. Conf. on Communication Tech. (ICCT), pp. 270-273, October 2015.
- [9] T. Adiono, S. Fuada, A.P. Putra, Y. Aska, "Desain Awal Analog Front-End (AFE) Optical Transceiver untuk aplikasi Visible Light Communication (VLC)," unpublished.
- [10] T. Adiono, S. Fuada, A.P. Putra, Y. Aska, "Proposed A New Analog Front-End Transceiver Design for OFDM Visible Light Communication," unpublished.
- [11] A. Bhat, "Stabilize Your Transimpedance Amplifier," Application Note 5129, February 2012.
- [12] Datasheet of photodiode SP8-ML Kodenshi, [Online], Available at: <http://www.datasheetarchive.kr/dl/566f3361a9f33a0d722a89e8ce168132f7ed17/O/SP8ML>.
- [13] S. Fuada, T. Adiono, Y. Aska, A.P. Putra, "Noise Analysis in VLC Optical Link based Discrete OP-AMP Trans-Impedance Amplifier (TIA)," unpublished.
- [14] M. Nakhostin, *et al.*, "Use of commercial operational amplifiers in a low cost multi-channel preamplifier system," Radiation Physics and Chemistry, Vol. 85, pp. 18-22, 2013.
- [15] B. Mishra, K. Sharma, P. Choudhary, "A Low Noise Op-Amp Transimpedance Amplifier for InGaAs Photodetectors," Int. Research J. of Engg. And Tech. (IRJET), Vol.2(5), pp. 54-58, Agust 2015.
- [16] A. P. Alexander Weiler, "High-speed layout guidelines," Texas Instrumensts, Tech. Rep., November 2006. [Online]. Available: <http://www.ti.com/lit/an/scaa082/scaa082.pdf>.
- [17] J. Caldwell, "1 MHz, Single-Supply, Photodiode Amplifier Reference Design," TI Application Note TIDU535, pp. 1-19, November 2014.
- [18] W. Wang, *et al.*, "A Pre-amplifier Circuit for VLC System," Proc. of the IEE 6th Int. Sym. On Microwave, Antenna, Propagation, and EMC Technologies (MAPE),pp. 686-688, October 2015.
- [19] A. Pradana, "Rancang Bangun Layer Fisik Komunikasi Cahaya Tampak Berbasis DC-OFDM dan PWM," Master Thesis, Institut Teknologi Bandung (ITB), Indonesia, Agust 2016.
- [20] A. Assabir, J. Elmhamdi, A. Hammouch, A. Akherraz, "Application of Li-Fi Technology in the Transmission of the Sound at the Base of the PWM," Proc. of the 2nd Int. Conf. on Electrical and Information Technologies (ICEIT), May 2016, DOI: [10.1109/EITech.2016.7519602](https://doi.org/10.1109/EITech.2016.7519602).
- [21] L. Orozco, "Programmable-Gain Transimpedance Amplifiers Maximize Dynamic Range in Spectroscopy Systems, Analog Dialoue, Vol. 47, May 2013. [Online] Available at: <http://www.analog.com/library/analogdialogue/archives/47-05/pgtia.html>.

